

RL Reasoning for Recommendation Ranking

ABSTRACT

We propose a framework that applies DeepSeek-R1-style GRPO to train LLMs as reasoning-capable recommendation rankers. Existing LLM-based rankers rely on zero-shot prompting or SFT, both of which fail to align the model with listwise ranking objectives. Our method introduces rank-level reward decomposition and a chain-of-thought reasoning trace to enable verifiable, position-aware RL training without proprietary data or human annotations.

TL;DR: An RL framework that teaches LLMs to reason before ranking, using rank-level DCG rewards as verifiable signals — no GPT-4o, no human labels.

§1.2 Motivation

Why is this interesting?

- LLMs' ranking quality **degrades sharply toward lower list positions** — a consequence of sequence-level objectives misaligned with NDCG.
- SFT-based rankers catastrophically forget general capabilities — GSM8K drops 28 pts, IFEval drops 27 pts after SFT [Lin et al., 2025].
- RLVR has proved reasoning can emerge from pure RL without human labels [DeepSeek-AI, 2025]. Can the same principle unlock better ranking?

Prior works & state-of-the-art

Recent (last 2 years): [Lin et al., 2025], [Zhu et al., 2025], [Hou et al., 2024], [DeepSeek-AI, 2025], [FlexRec, 2025]

Seminal: [Hu et al., 2018], [Shao et al., 2024], [Hou et al., 2023]

State-of-the-art: ConvRec-R1 / Rank-GRPO [Zhu et al., 2025] — rank-level GRPO with DCG-weighted rewards, ICLR 2026.

CHALLENGE 1

Non-causal credit assignment. Vanilla GRPO assigns one scalar reward per sequence. For ranked lists, only top-K positions drive NDCG, but back-prop spreads the gradient uniformly — ranks 1 and 5 receive identical updates.

CHALLENGE 2

Verifiable reward design. Unlike math tasks with exact answers, recommendation quality is relative to user preferences. Defining a clean, per-step verifiable reward without human annotation is non-trivial.

CHALLENGE 3

Catalog grounding. Open-vocabulary LLMs frequently hallucinate item IDs absent from the catalog. A warm-start pipeline that avoids dependency on proprietary models (e.g., GPT-4o) is needed.

§2 Problem Formulation

We consider an **episodic MDP** $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{R}, \gamma, \pi_\phi)$ for LLM-based recommendation ranking.

Component	Definition
State $s \in \mathcal{S}$	User query + sequential interaction history + top- K candidate list retrieved via BM25 or dense retriever
Action $a \in \mathcal{A}$	A ranked list of N items generated by LLM policy π_ϕ , optionally prefixed with a chain-of-thought trace inside <code><think>...</think></code>
Reward $\mathcal{R}(s, a)$	NDCG@ K or Recall@ K computed deterministically against held-out interactions; format reward $+\varepsilon$ if <code><think></code> block is present and list parses correctly
Policy π_ϕ	Pre-trained LLM (Qwen2.5-0.5B or Llama3.2-3B) fine-tuned via (1) SFT warm-start, then (2) GRPO with rank-level rewards
Objective	$\max_{\pi \in \Pi} \mathbb{E}_{s \sim \mathcal{D}, a \sim \pi} [\mathcal{R}(s, a)]$
Assumptions	Catalog is finite and known at inference time. Relevance labels exist for offline evaluation. Episodic MDP — no cross-session dependency.

§3 Empirical Evaluation

§3.1 Performance Metrics

NDCG@K

Primary listwise ranking quality; position-weighted and directly tied to our reward signal.

RECALL@K

Proportion of relevant items in top-K; measures coverage of user-preferred items.

WALL-CLOCK TIME

Training efficiency per epoch on the same GPU (A100); measures computational overhead vs. baselines.

CATALOG HIT RATE

% of ranked items existing in the catalog at evaluation; monitors hallucination rate.

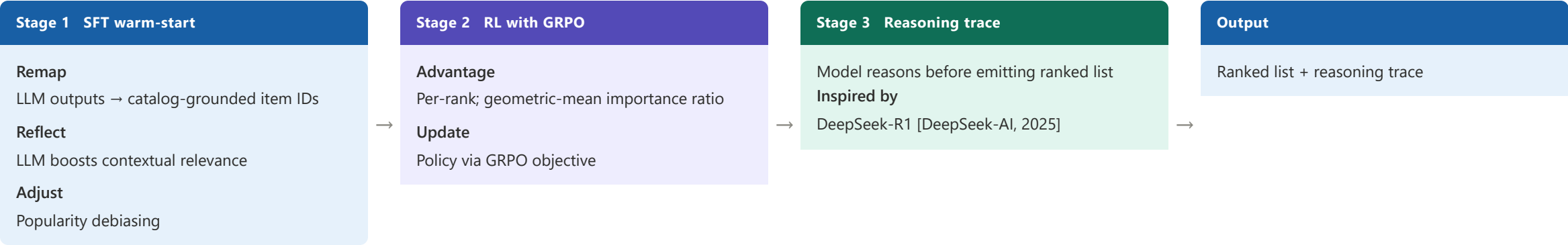
§3.2 Baseline Methods

- **SFT-only** (ablation): Supervised fine-tuning on behavioral cloning data only; measures the gain of adding RL on top of SFT.
- **LLMRank** [Hou et al., 2024]: Zero-shot LLM ranker via instruction prompting; no RL training. Establishes the prompting baseline. github.com/RUCAIBox/LLMRank
- **ConvRec-R1** [Zhu et al., 2025]: Rank-level GRPO (Rank-GRPO) with DCG position-weighted rewards and SFT warm-start. Current SOTA on Reddit-v2. github.com/yaochenzhu/Rank-GRPO

§3.3 Benchmark Tasks / Datasets

- **MovieLens-1M** (1M interactions): Sequential recommendation; standard LLM4Rec benchmark [Hou et al., 2024].
- **Reddit-v2** (~400K conversations): Primary conversational recommendation benchmark; used in [Zhu et al., 2025].

§4 Methodology



Reward decomposition (rank-level GRPO)



References

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